Shangwen Sun

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Education

Baruch College, City University of New York New York, NY
Master of Science in Financial Engineering 08/2021 – 12/2022
Selected Courses: Machine Learning, Natural Language Processing, Stochastic Processes

School of Mathematical Sciences, Peking University Beijing, China
Bachelor of Science in Mathematics and Applied Mathematics 09/2016 – 07/2021
Bachelor of Economics Minor
Selected Courses: Differential Equations, Advanced Algebra, Bayesian Theory

Work Experience

New York University New York, NY
Research Assistant 04/2025 – Present
Hosted by Prof. Alfredo Canziani and Prof. Yann LeCun

- Conducted exploratory experiments on open-source LLM models, investigating mechanisms underlying massive activations and attention sinks.
- Wrote PyTorch code for multi-node LLM training using FSDP to run ablation studies on Llama-style architectures.

Millennium Management New York, NY
Quantitative Researcher, Model-Driven Algorithmic Trading Team 01/2023 – 03/2025
Quantitative Developer Intern, Statistical Arbitrage Platform 06/2022 – 08/2022

- Conducted research on U.S. equity order execution and built batch-parallelized research tools, enabling faster tick-level experiments and more efficient strategy evaluation.
- Designed and deployed short-horizon signals for an in-house VWAP algorithm, reducing execution costs on targeted order flows, and built post-trade analysis models that incorporated historical flow features to assess broker performance.
- Performed P&L attribution and seasonal portfolio analysis across customized risk factors and volatility regimes, and corrected errors in an in-house Barra risk model.

Publications

Anatomy of Massive Activations and Attention Sinks *Under Review*
Shangwen Sun, Alfredo Canziani, Yann LeCun, Jiachen Zhu
We provide a mechanistic account of why massive activations and attention sinks co-emerge in pretrained LLMs. Key insights: normalization is the main architectural bridge, converting massive activations into near-constant hidden states that can induce sinks; sink behavior is shaped by the geometric properties of the attention-space and the training context-length distribution, helping bias some heads toward short-range structure; and the two phenomena can be suppressed independently without hurting modeling performances, suggesting their overlap is an incidental consequence of current design choices rather than a functional necessity.

Activities

Vice President, European Society, Peking University 2017 – 2018

Selected Awards

Meritorious Winner, The Mathematical Contest in Modeling
Gold Medal, 14th Chinese Girls Mathematical Olympiad
Silver Medal, 31st Chinese Mathematical Olympiad